LINUX FOR DEVOPS

Field Notes: Enough for DevOps

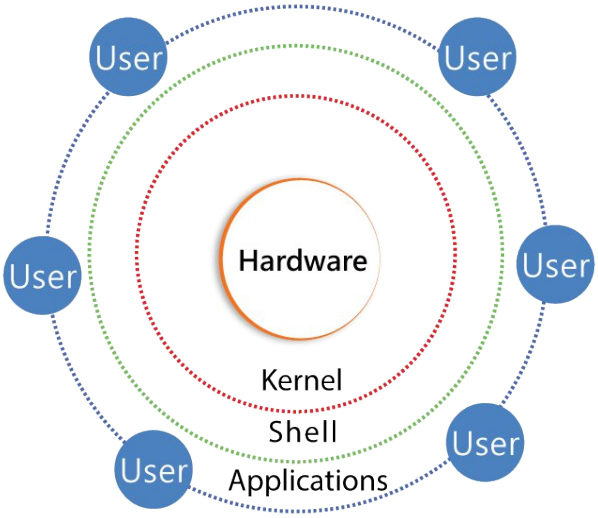
What this covers

- → Linux architecture, processes & systemd
- → File system hierarchy & navigation
- → Permissions, ownership & access control
- → Process management & troubleshooting
- → Networking, service management & logs
- → LVM storage management
- → SSH access & server-to-server connectivity

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➢ Linux Architecture



User Applications --> Shell (bash/zsh) —>System Calls —>Kernel (Process, Memory, File, Network Mgmt) —>Hardware (CPU, RAM, Disk, NIC)

Kernel — core jo hardware ko manage karta hai (scheduling, memory, drivers)

Shell — user aur kernel ke beech interface (commands leta hai)

System calls — shell/programs kernel se baat karte hain inke through (e.g., `open()`, `fork()`, `read()`)

➢ Linux file system hierarchy

Directory	Purpose
/	Root directory
/etc	Configuration files
/home	User directories

/var	Logs and variable data
/tmp	Temporary files
/bin	Essential commands
/usr	Installed programs
/boot	Boot files
/dev	Hardware devices
/opt	Optional software
/root	Root user's home
/proc	Process information

Note: Don’t forget below rule

Configuration → /etc

User files → /home

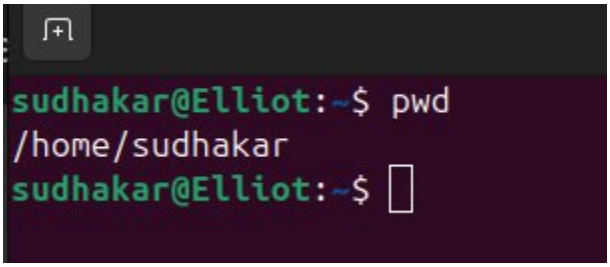
Logs → /var/log

Temporary files → /tmp

Programs → /usr/bin

➢ Navigate the file system, create/move/delete files and directories

pwd – print the current directory

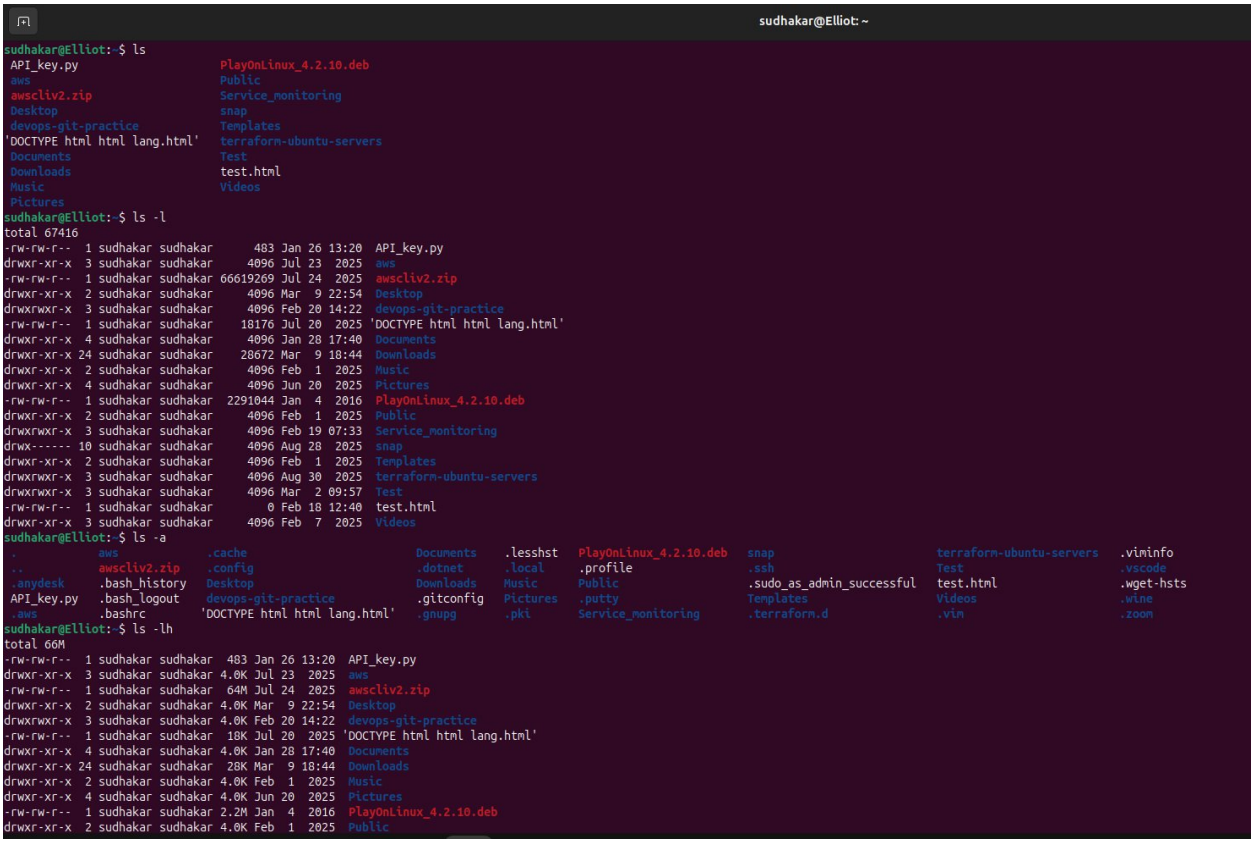


ls – list file and directory

ls –l — print file and directory in long format

ls –a — show hidden file

ls –lh — show human readable file size



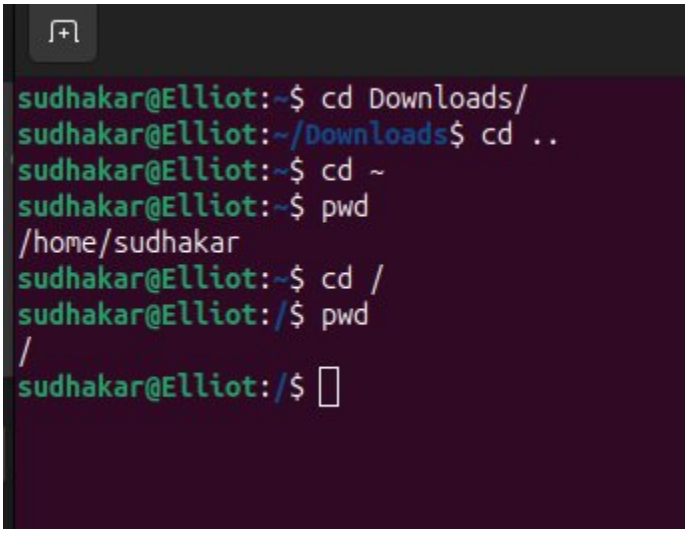
Cd Change Directory

cd <file name> – move into folder

cd .. – move back one directory

cd ~ – go to home direc

cd / – go to root direc

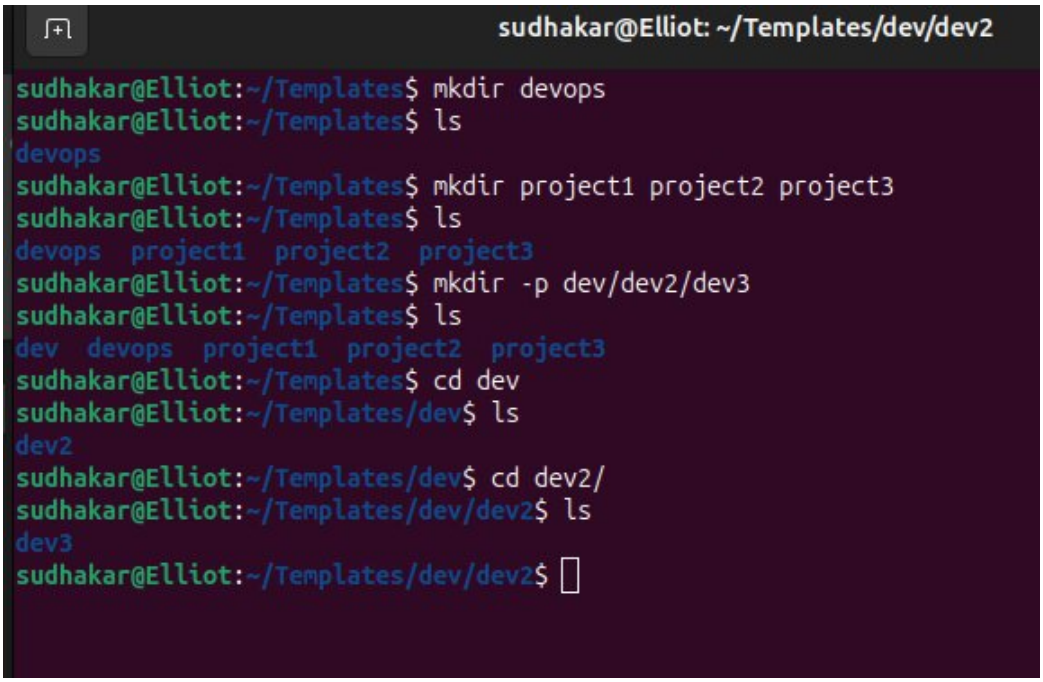


Create File and directory

mkdir <file name>- create directory

mkdir project1 project 2 – create multiple dirc

mkdir –p <fold>/<fold2>/<fold3> – nested direc

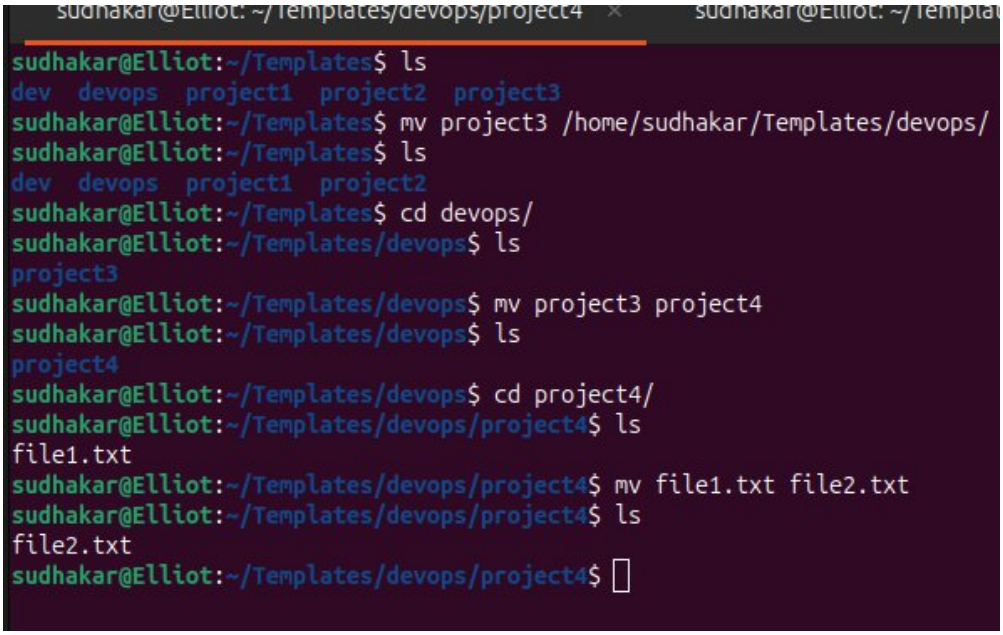


Move and Rename files

mv project3 /home/sudhakar/templates/project2 – move and rename

mv project3 project4 – move directory

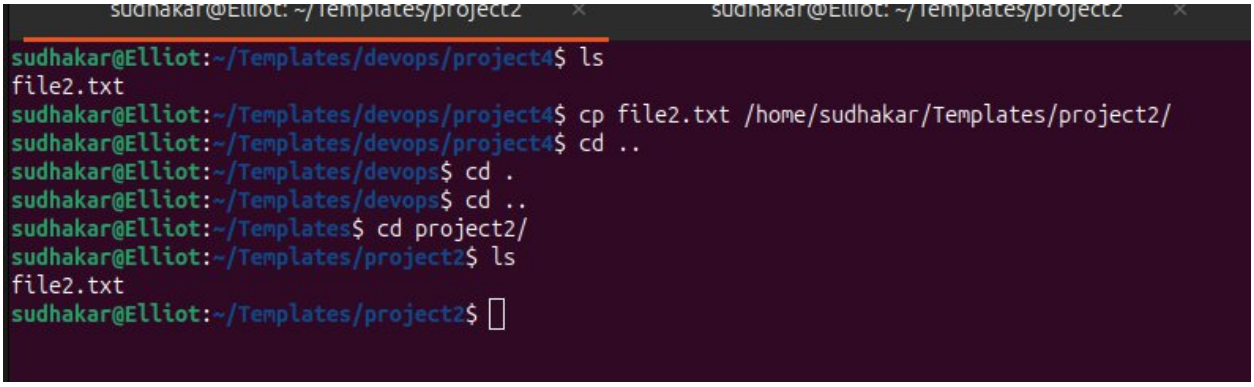
mv file1.txt file2.txt – Rename the file



Copy file and directory

cp copy file and directory

cp <file name> <destination>



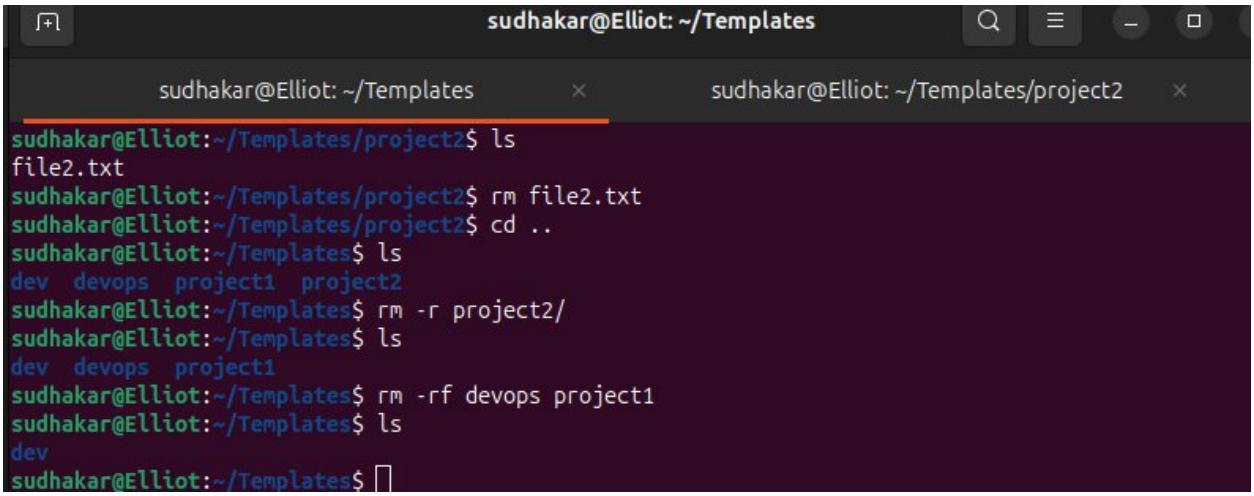
Deleting file and directory

rm file1.txt – remove file

rm file1.txt file2txt file3.txt – remove multiple file

rm –r devops2 – remove directory

rm –rf devops1 devops1 – remove forcefully directory



Understanding Linux permissions

Every file have three permissions group

User (u) – owner of the file

Group (g) –user in this file group

Other (o) – everyone else

And three types of permission

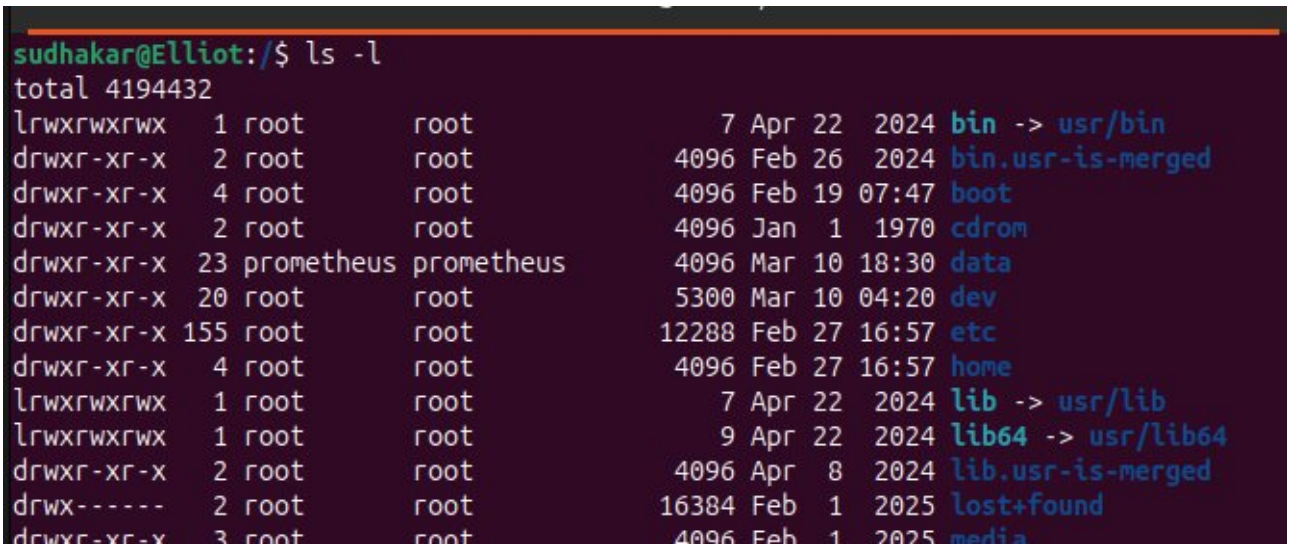
Read r view all context

Write w modify file

Execute x run file as a program

Check file permissions

ls –l



chmod 755 file1.sh – change permission file

chmod +x script.sh – Add and execute permission

chmod u-w file.txt – remove writen permission

chmod g+r file.txt – give read permission

Change file ownership with chown and chgrp

ls -l

-rw-r--r-- 1 sudhakar devops 1200 Mar 10 script.sh

```
          |          |
          |          └── Group (devops)
          └──────────── Owner (sudhakar)
```

chown suraj file,txt

chown suraj project – change the ownership of Directory

chown –R suraj project – (recursive ownership) to change the ownership of a folder and every thing inside it

Change group

chgrp <groupname> file

➢ Manage processes — list, kill, background/foreground

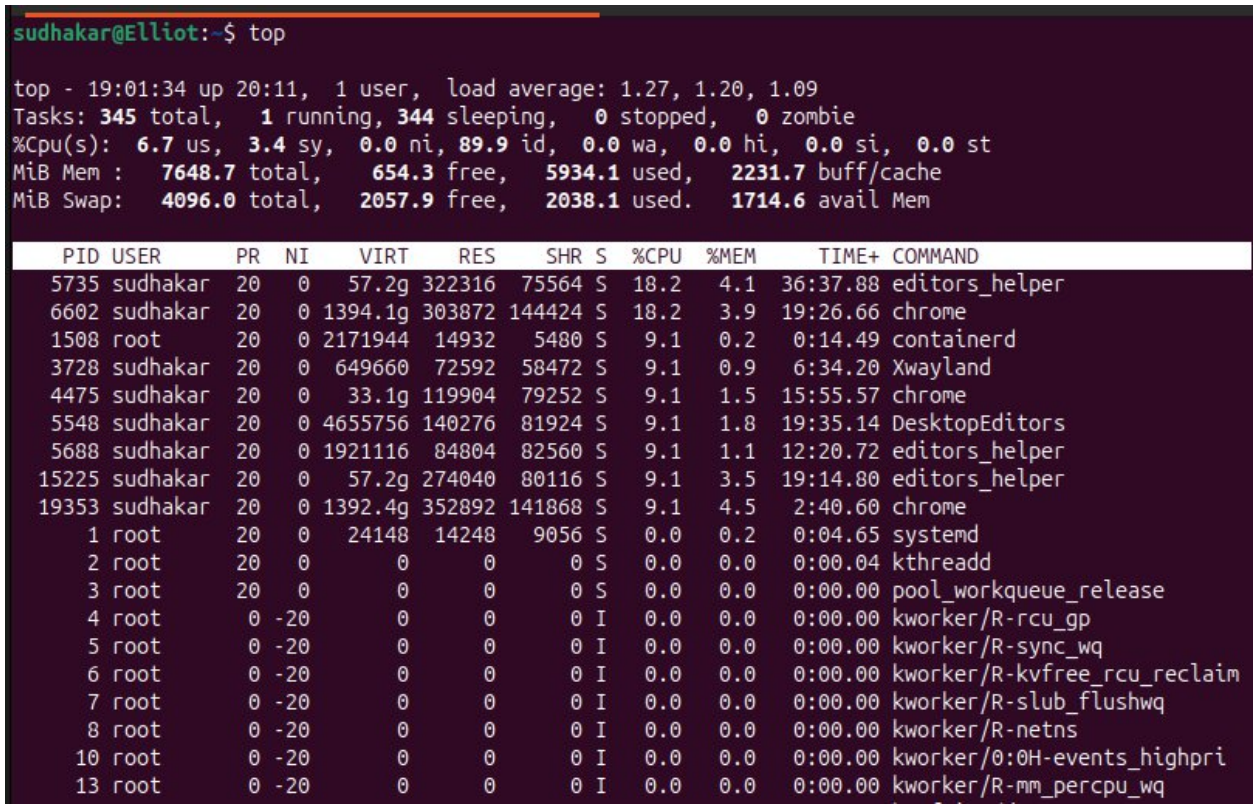
List running status

ps – process status

ps aux – show all process

```
sudhakar@Elliot:~$ ps
    PID TTY          TIME CMD
   7266 pts/0    00:00:00 bash
  25344 pts/0    00:00:00 ps
sudhakar@Elliot:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24148 14248 ?        Ss   Mar09   0:04 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    Mar09   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    Mar09   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/R-kvfree_rcu_recl
root           7  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/0:0H-events_highp
root          13  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    Mar09   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    Mar09   0:42 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    Mar09   0:00 [rcu_exp_par_gp_kthread_wo
root          17  0.0  0.0      0     0 ?        S    Mar09   0:00 [rcu_exp_gp_kthread_worker
root          18  0.0  0.0      0     0 ?        S    Mar09   0:05 [migration/0]
root          19  0.0  0.0      0     0 ?        S    Mar09   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    Mar09   0:00 [cpuhp/0]
root          21  0.0  0.0      0     0 ?        S    Mar09   0:00 [cpuhp/2]
root          22  0.0  0.0      0     0 ?        S    Mar09   0:00 [idle_inject/2]
root          23  0.0  0.0      0     0 ?        S    Mar09   0:04 [migration/2]
root          24  0.0  0.0      0     0 ?        S    Mar09   0:00 [ksoftirqd/2]
root          26  0.0  0.0      0     0 ?        I<   Mar09   0:00 [kworker/2:0H-events_highp
```

Top – for real time status



ps aux | grep <process ID> – find process

Kill <process ID> – kill a process

Kill -9 <process ID > – kill a process forcefully



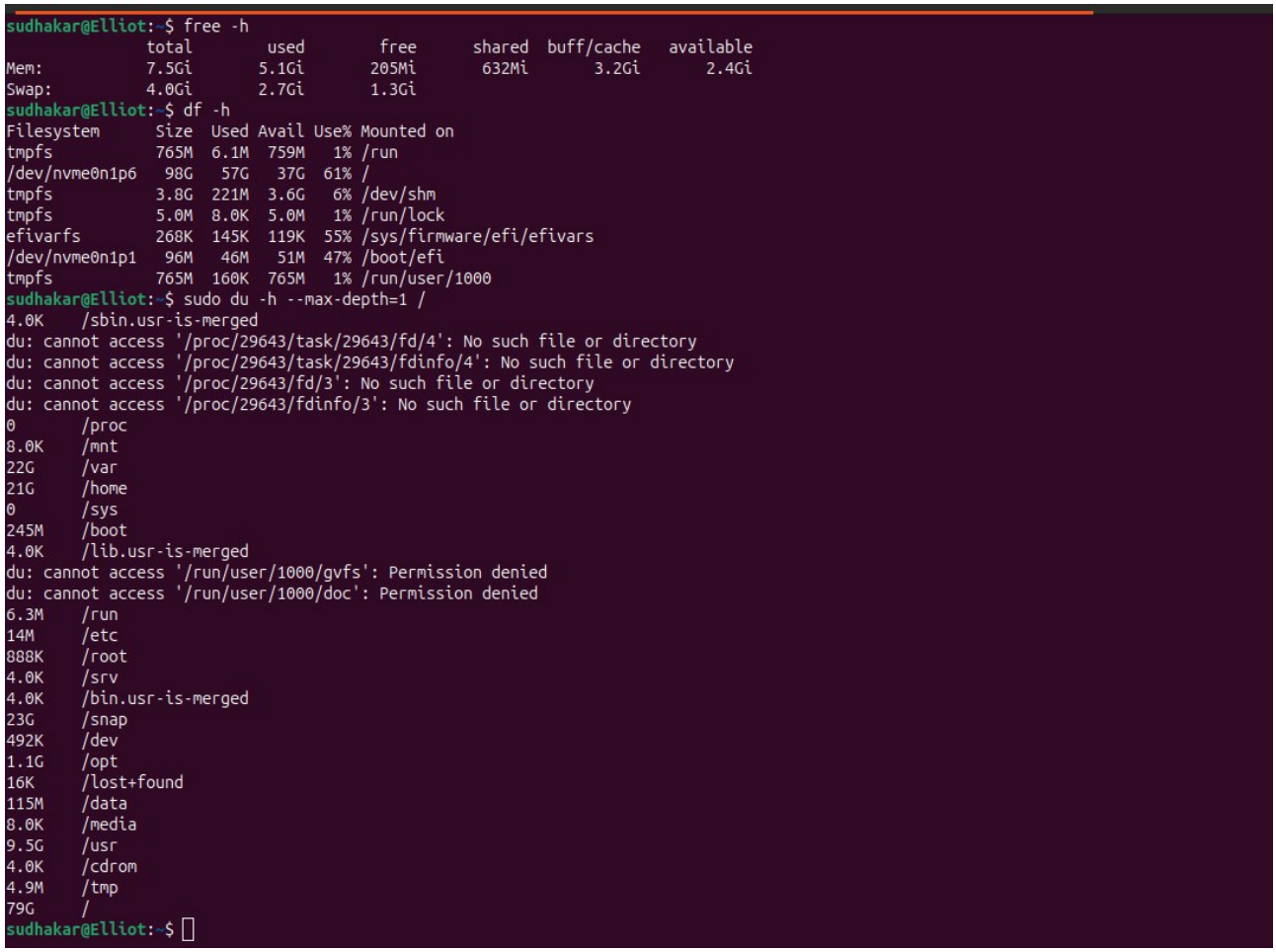
➢ Troubleshoot CPU, memory, and disk issues using top, free, df, du

top – to check hight process

free -h - To check memory uses

df -h – check disk

du -h --max-depth=1 / – find large directory



➢ Check network connectivity — ping, curl, netstat, ss, dig, nslookup

When troubleshooting networks in Linux, you are basically asking three detective questions

- Is the server reachable?
- Is the service responding?
- Is DNS resolving correctly?

several commands for this: ping, curl, netstat, ss, dig, and nslookup

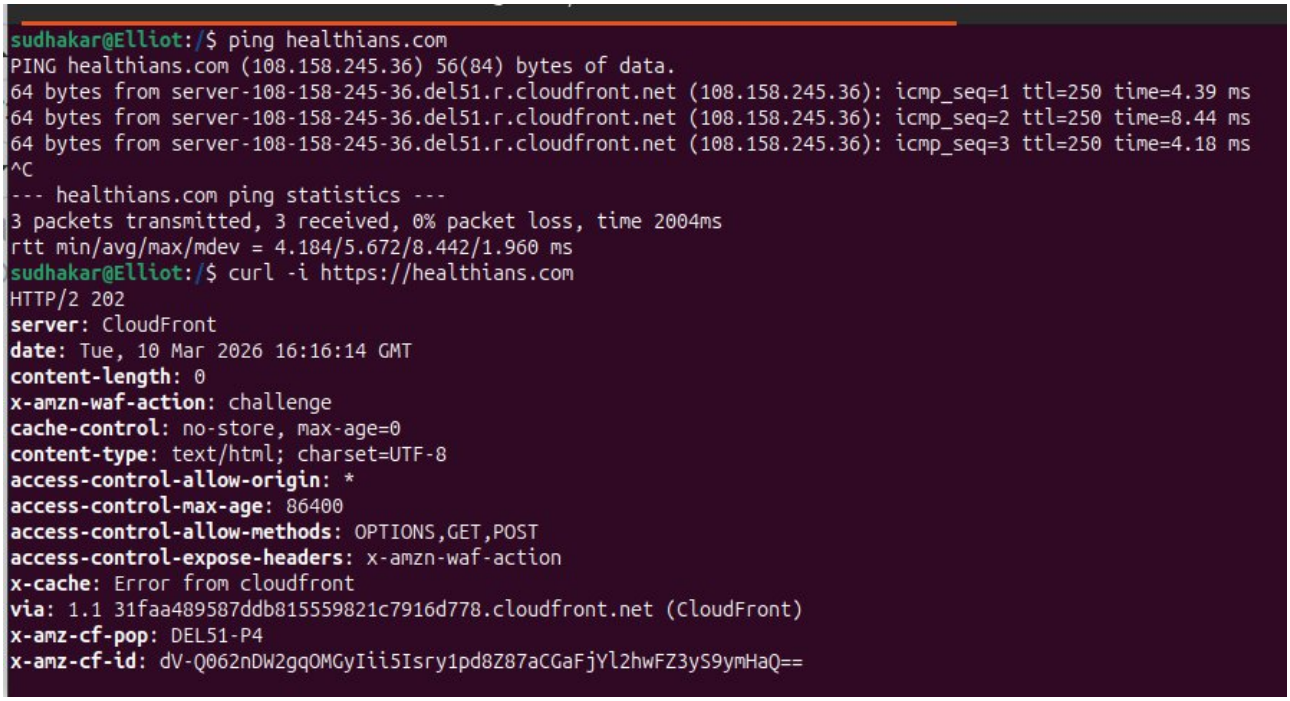
ping google.com – check basic network connectivity

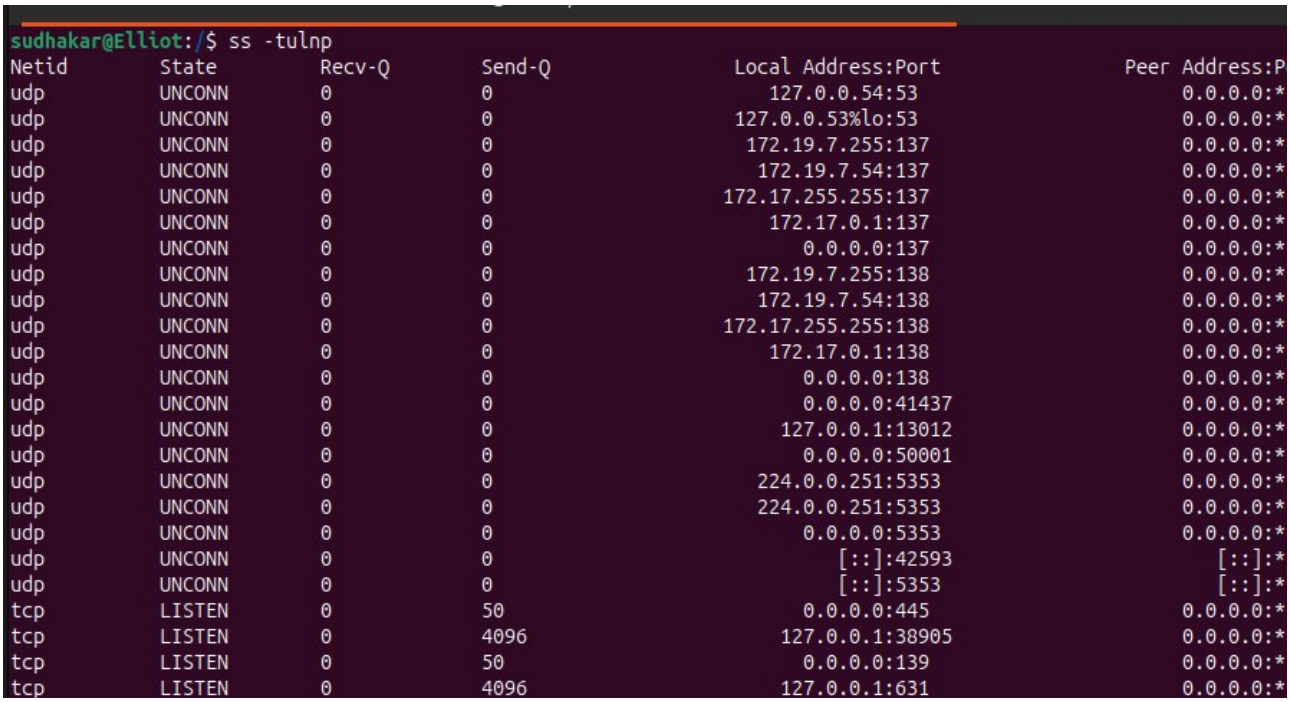
curl https://healthians.com – test web service/API

ss –tulnp – show network connection

dig google.com – DND lookup

nslookup google.com – simple DND lookup





```
sudhakar@Elliot:/$ ss -tulnp
Netid       State        Recv-Q       Send-Q             Local Address:Port              Peer Address:P
udp         UNCONN       0            0                     127.0.0.54:53                   0.0.0.0:*
udp         UNCONN       0            0                  127.0.0.53%lo:53                   0.0.0.0:*
udp         UNCONN       0            0                  172.19.7.255:137                   0.0.0.0:*
udp         UNCONN       0            0                   172.19.7.54:137                   0.0.0.0:*
udp         UNCONN       0            0                172.17.255.255:137                   0.0.0.0:*
udp         UNCONN       0            0                    172.17.0.1:137                   0.0.0.0:*
udp         UNCONN       0            0                       0.0.0.0:137                   0.0.0.0:*
udp         UNCONN       0            0                  172.19.7.255:138                   0.0.0.0:*
udp         UNCONN       0            0                   172.19.7.54:138                   0.0.0.0:*
udp         UNCONN       0            0                172.17.255.255:138                   0.0.0.0:*
udp         UNCONN       0            0                    172.17.0.1:138                   0.0.0.0:*
udp         UNCONN       0            0                       0.0.0.0:138                   0.0.0.0:*
udp         UNCONN       0            0                       0.0.0.0:41437                 0.0.0.0:*
udp         UNCONN       0            0                    127.0.0.1:13012                  0.0.0.0:*
udp         UNCONN       0            0                       0.0.0.0:50001                 0.0.0.0:*
udp         UNCONN       0            0                  224.0.0.251:5353                    0.0.0.0:*
udp         UNCONN       0            0                  224.0.0.251:5353                    0.0.0.0:*
udp         UNCONN       0            0                       0.0.0.0:5353                  0.0.0.0:*
udp         UNCONN       0            0                          [::]:42593                    [::]:*
udp         UNCONN       0            0                          [::]:5353                     [::]:*
tcp         LISTEN       0            50                      0.0.0.0:445                   0.0.0.0:*
tcp         LISTEN       0            4096                 127.0.0.1:38905                  0.0.0.0:*
tcp         LISTEN       0            50                      0.0.0.0:139                   0.0.0.0:*
tcp         LISTEN       0            4096                 127.0.0.1:631                    0.0.0.0:*
```

This shows:

- active ports
- listening services
- process IDs

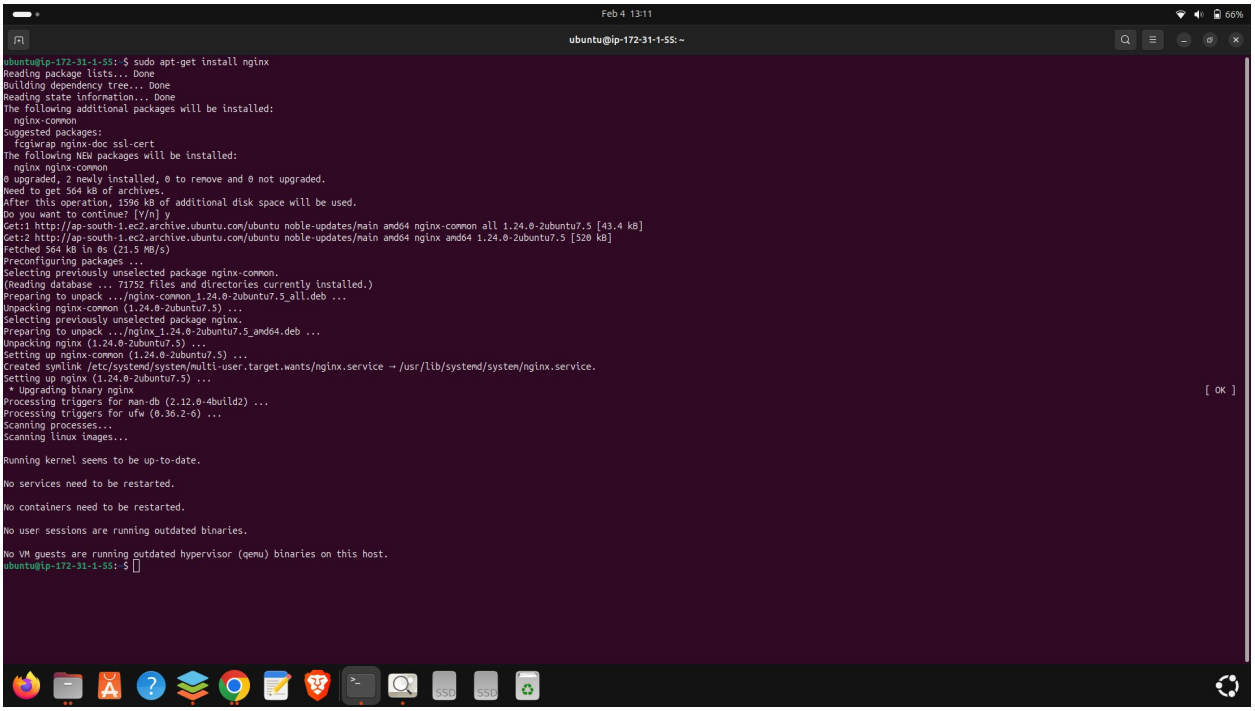
Some Common but important PORTS

Port	Service
20	FTP data
21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
143	IMAP
443	HTTPS

3306	MySQL
5432	PostgreSQL
6379	Redis
8080	Web apps

➢ Install of any program and checking Status and logs:-

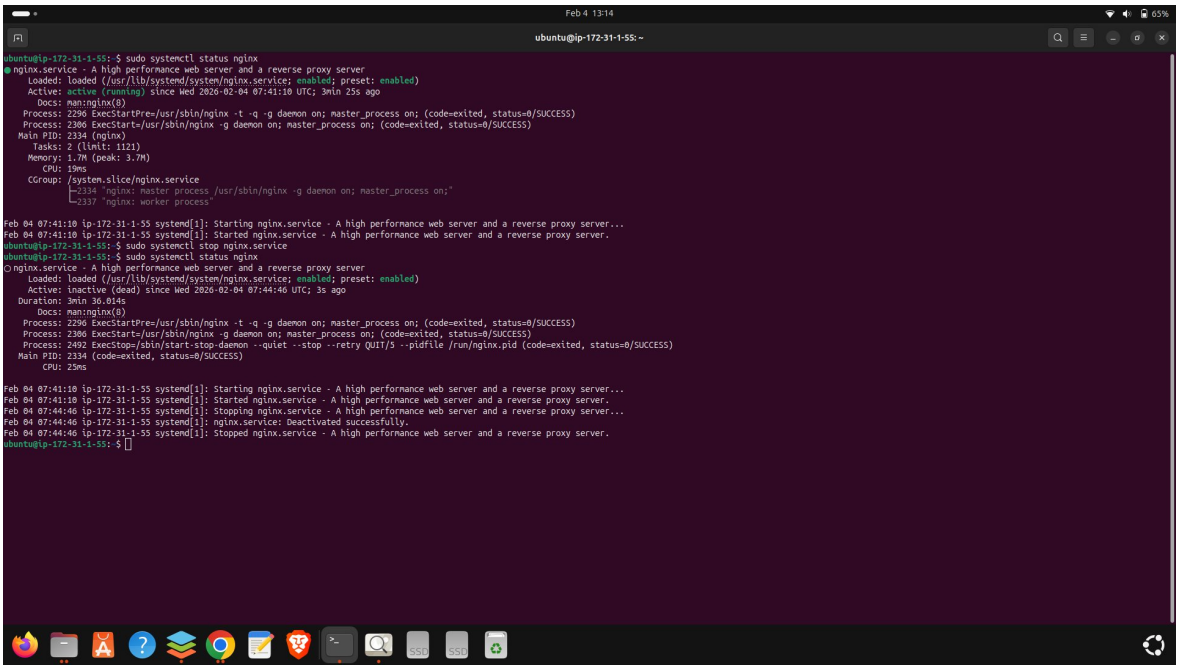
$ Sudo apt-get install nginx



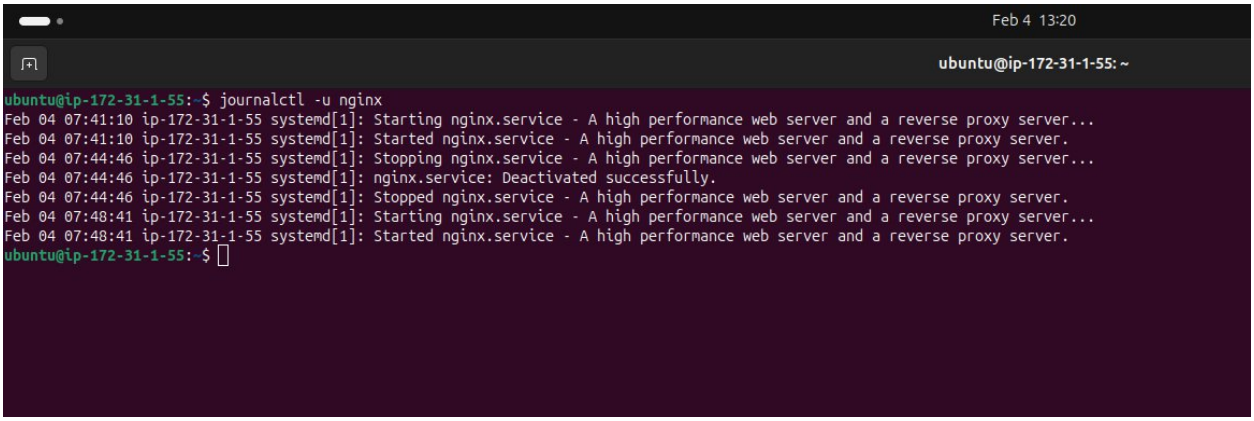
$ Systemctl status nginx ->> to check status of any program

$ systemctl stop nginx ->> to stop any program

$ systemctl start nginx ->> to start any program



$ journalctl –u nginx —>>> to check logs of the process



➢ (L V M) Logical Volume Management:-

LVM is a storage management layer in Linux that lets you:

- Combine multiple disks
- Resize storage without downtime
- Manage disks flexibly, not rigidly

Disk → Physical Vol → Volume Group → Logical Vol → Filesystem → Mount

Environment Details (Lab Setup)

- OS: Ubuntu (AWS EC2)
- User: root
- Disks attached:

  `/dev/xvdf` → 10 GB

  `/dev/xvdg` → 15 GB

- Volume Group: `vg_data`
- Logical Volume: `lv_data`
- Mount Point: `/data1`

$Sudo su

$lsblk

lvm

lvm> pvcreate /dev/xvdf

lvm> pvcreate /dev/xvdg

lvm> vgcreate vg_data /dev/xvdf

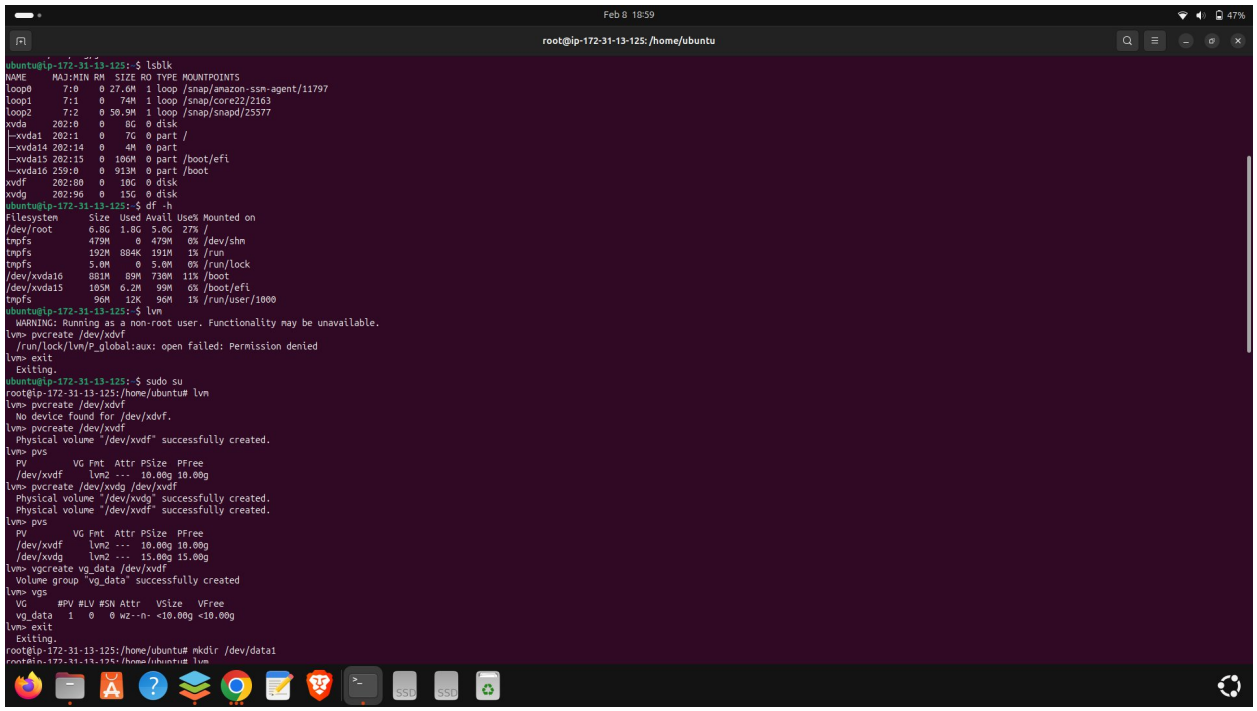
lvm> lvcreate -L 5G -n lv_data vg_data

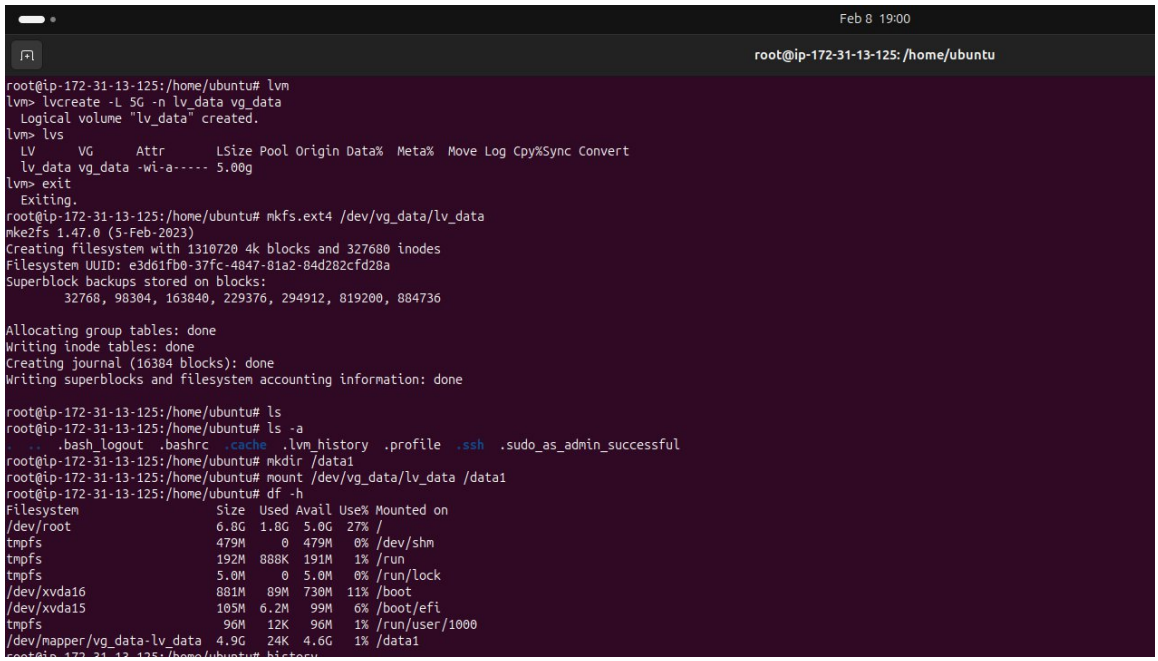
Exit

$ mkfs.ext4 /dev/vg_data/lv_data

$ mkdir /data1

$ mount /dev/vg_data/lv_data /data1


Feb 8 18:59
47%
root@ip-172-31-13-125: /home/ubuntu
ubuntu@ip-172-31-13-125: $ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1 7:1 0 74M 1 loop /snap/core22/2163
loop2 7:2 0 50.9M 1 loop /snap/snapd/25577
xvda 202:0 0 8G 0 disk
├─xvda1 202:1 0 7G 0 part /
├─xvda14 202:14 0 4M 0 part
├─xvda15 202:15 0 106M 0 part /boot/efi
└─xvda16 259:0 0 913M 0 part /boot
xvdf 202:80 0 10G 0 disk
xvdg 202:96 0 15G 0 disk
ubuntu@ip-172-31-13-125: $ df -h
Filesystem Size Used Avail Use% Mounted on
/dev/root 6.8G 1.8G 5.0G 27% /
tmpfs 479M 0 479M 0% /dev/shm
tmpfs 192M 884K 191M 1% /run
tmpfs 5.0M 0 5.0M 0% /run/lock
/dev/xvda16 881M 89M 730M 11% /boot
/dev/xvda15 105M 6.2M 99M 6% /boot/efi
tmpfs 96M 12K 96M 1% /run/user/1000
ubuntu@ip-172-31-13-125: $ lvm
WARNING: Running as a non-root user. Functionality may be unavailable.
lvm> pvcreate /dev/xdvf
/run/lock/lvm/P_global:aux: open failed: Permission denied
lvm> exit
Exiting.
ubuntu@ip-172-31-13-125: $ sudo su
root@ip-172-31-13-125:/home/ubuntu# lvm
lvm> pvcreate /dev/xdvf
No device found for /dev/xdvf.
lvm> pvcreate /dev/xvdf
Physical volume "/dev/xvdf" successfully created.
lvm> pvs
PV VG Fmt Attr PSize PFree
/dev/xvdf lvm2 --- 10.00g 10.00g
lvm> pvcreate /dev/xvdg /dev/xvdf
Physical volume "/dev/xvdg" successfully created.
Physical volume "/dev/xvdf" successfully created.
lvm> pvs
PV VG Fmt Attr PSize PFree
/dev/xvdf lvm2 --- 10.00g 10.00g
/dev/xvdg lvm2 --- 15.00g 15.00g
lvm> vgcreate vg_data /dev/xvdf
Volume group "vg_data" successfully created
lvm> vgs
VG #PV #LV #SN Attr VSize VFree
vg_data 1 0 0 wz--n- <10.00g <10.00g
lvm> exit
Exiting.
root@ip-172-31-13-125:/home/ubuntu# mkdir /dev/data1

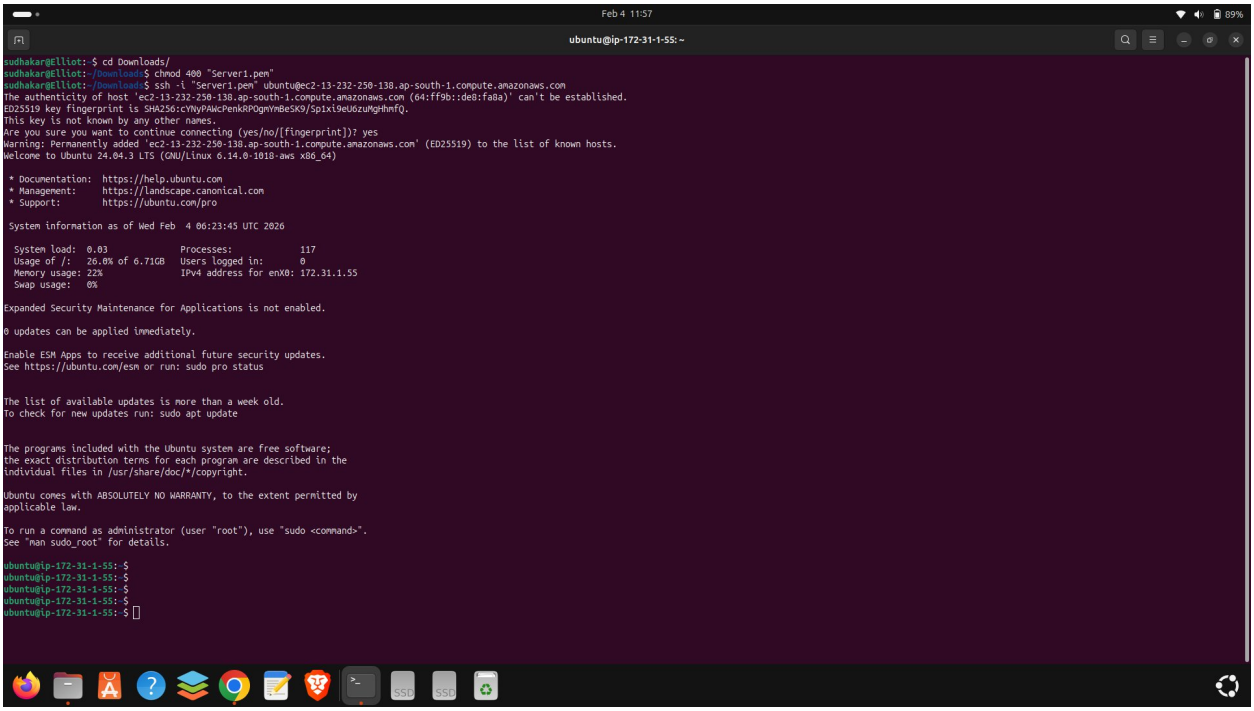

Feb 8 19:00
root@ip-172-31-13-125: /home/ubuntu
root@ip-172-31-13-125:/home/ubuntu# lvm
lvm> lvcreate -L 5G -n lv_data vg_data
Logical volume "lv_data" created.
lvm> lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
lv_data vg_data -wi-a----- 5.00g
lvm> exit
Exiting.
root@ip-172-31-13-125:/home/ubuntu# mkfs.ext4 /dev/vg_data/lv_data
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: e3d61fb0-37fc-4847-81a2-84d282cfd28a
Superblock backups stored on blocks:
32768, 98304, 163840, 229376, 294912, 819200, 884736
Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
root@ip-172-31-13-125:/home/ubuntu# ls
root@ip-172-31-13-125:/home/ubuntu# ls -a
. .. .bash_logout .bashrc .cache .lvm_history .profile .ssh .sudo_as_admin_successful
root@ip-172-31-13-125:/home/ubuntu# mkdir /data1
root@ip-172-31-13-125:/home/ubuntu# mount /dev/vg_data/lv_data /data1
root@ip-172-31-13-125:/home/ubuntu# df -h
Filesystem Size Used Avail Use% Mounted on
/dev/root 6.8G 1.8G 5.0G 27% /
tmpfs 479M 0 479M 0% /dev/shm
tmpfs 192M 888K 191M 1% /run
tmpfs 5.0M 0 5.0M 0% /run/lock
/dev/xvda16 881M 89M 730M 11% /boot
/dev/xvda15 105M 6.2M 99M 6% /boot/efi
tmpfs 96M 12K 96M 1% /run/user/1000
/dev/mapper/vg_data-lv_data 4.9G 24K 4.6G 1% /data1

➢ Create a EC2 instance and access from local system using SSH

$ cd Download/

$ chmod 400 “<private_Key-name>”

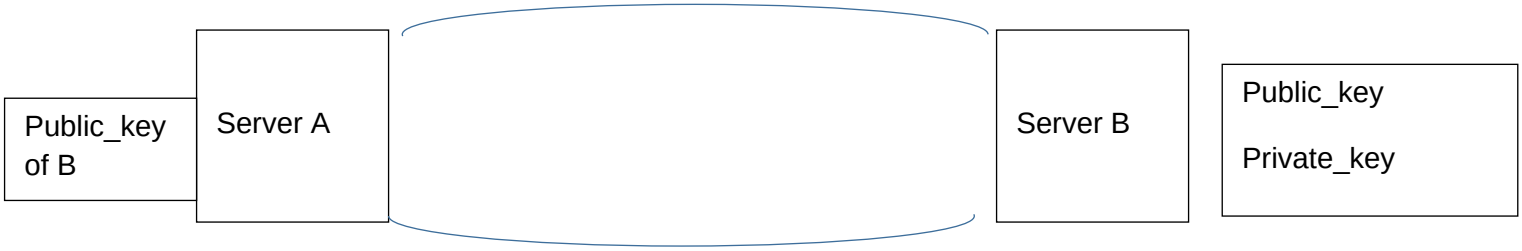
$ ssh –i “<private_key-name>” ubuntu@ec2-13-232-250-138.ap-south-1.compute.amazonaws.com



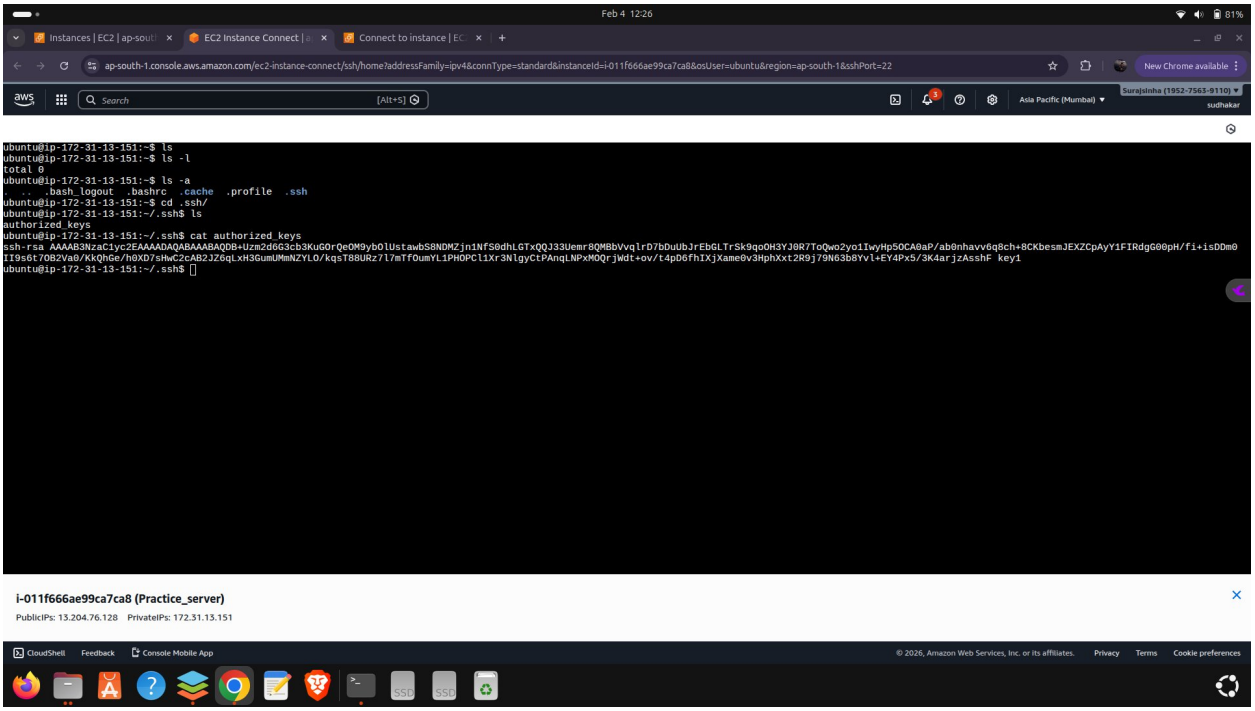
Now you are in your EC2 using SSH

Lets suppose you need to access another server using SSH

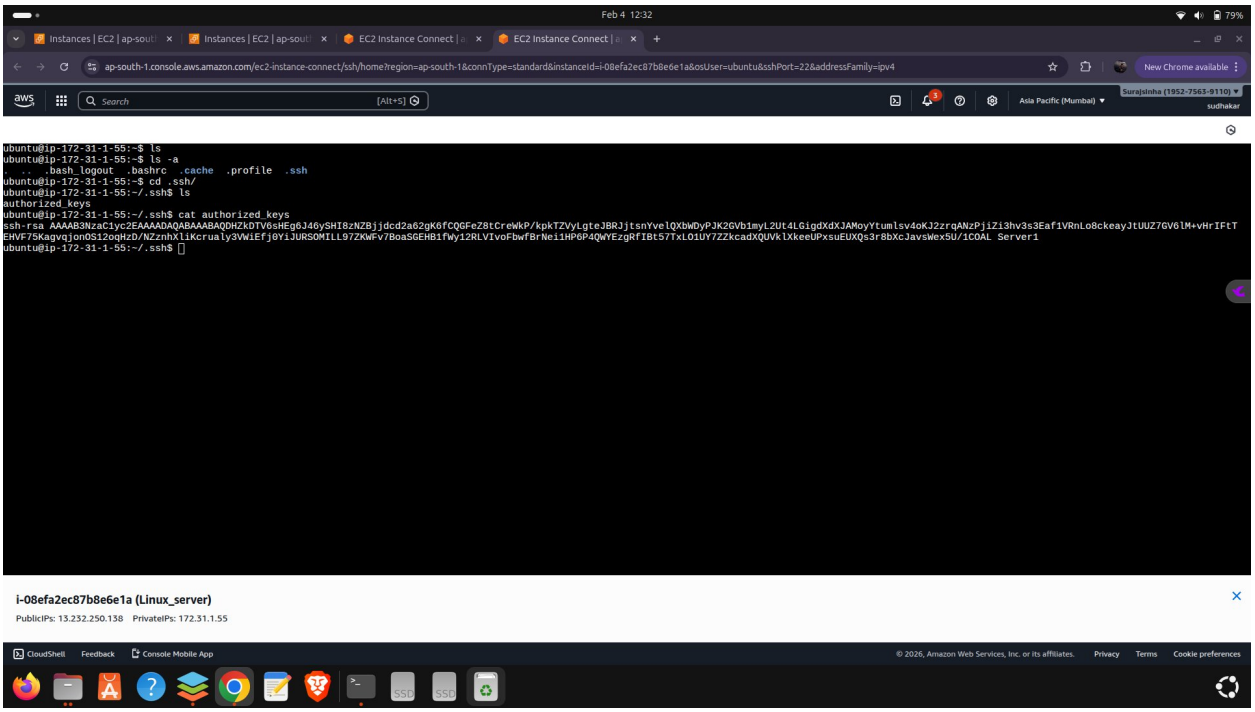
Here, you have server A and want to access through the server B so, you should have private_key of server B, and server B should have public_key



In this case I have server A (practice server) and server B (linux server)



Server B



Now, i am trying to access server A (Practice Server)

So, both server have there public_key in authorized_key here we need to generate the private_key and save in server A

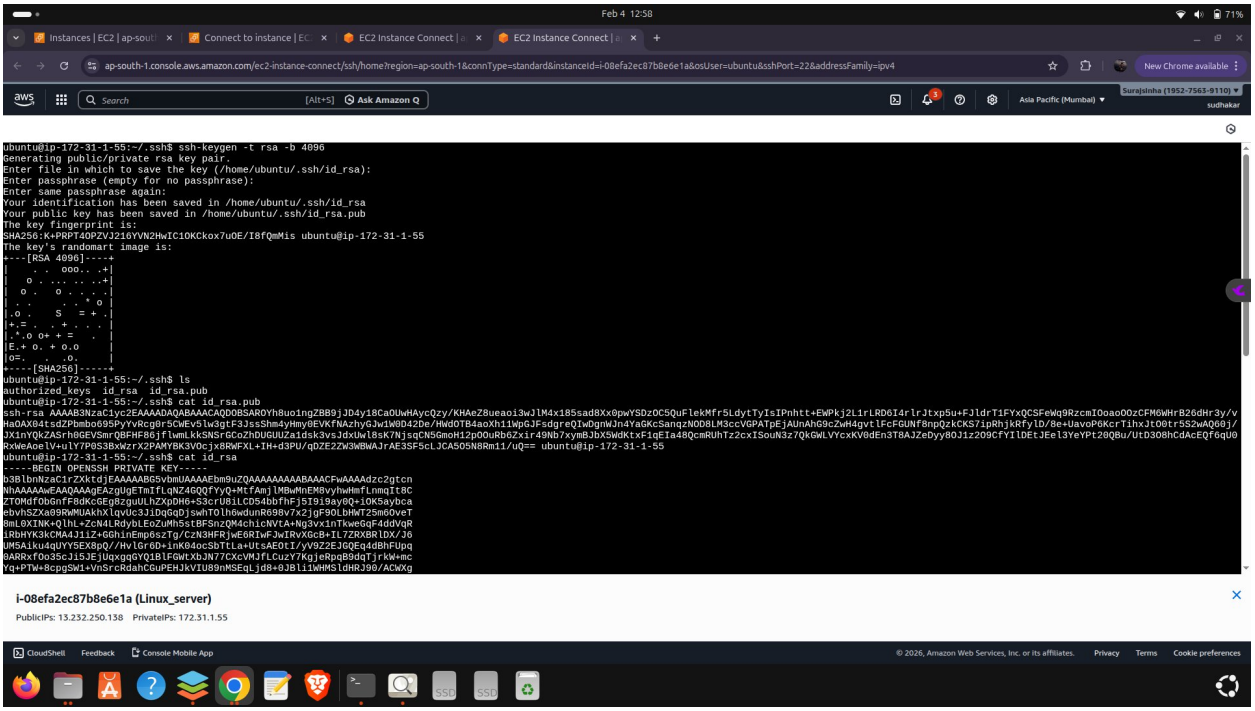
- Server A = Target EC2 (you want to login here)
- Server B = Source EC2 (you are SSH-ing from here)
- User = ubuntu

$ ssh-keygen -t rsa -b 4096

Key location: /home/ubuntu/.ssh/id_rsa

id_rsa → PRIVATE KEY × never share

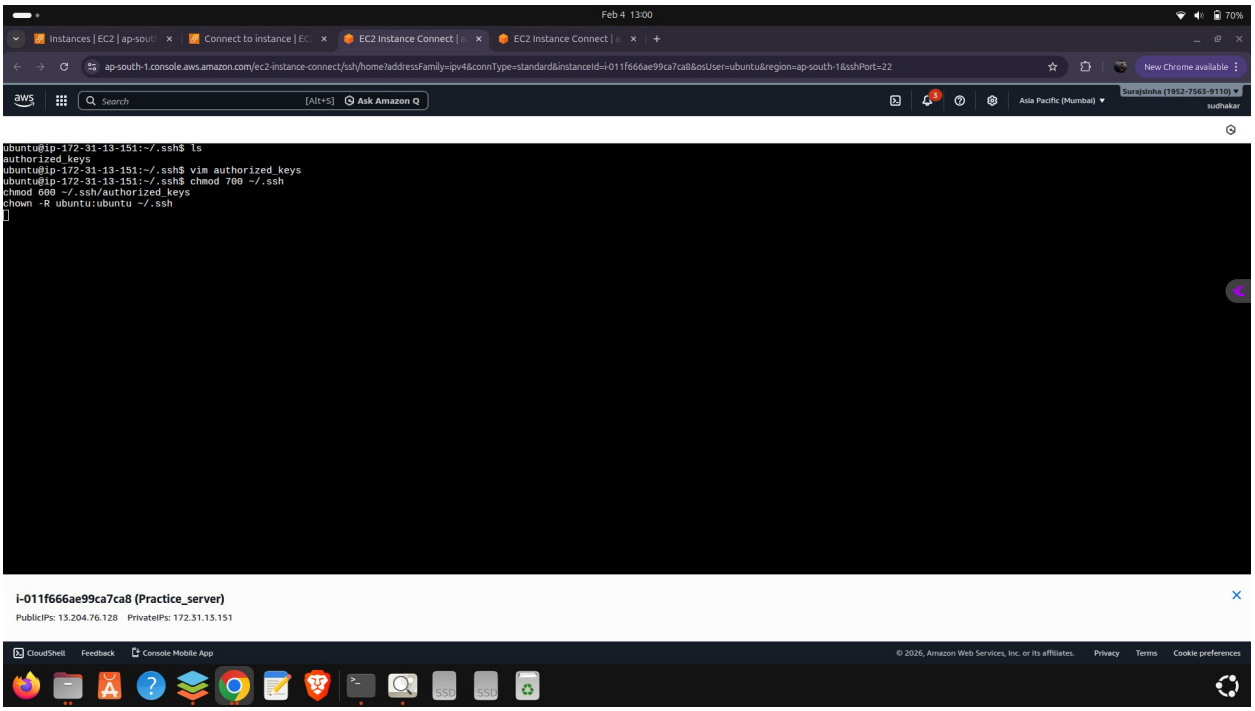
id_rsa.pub → PUBLIC KEY (Sharing)

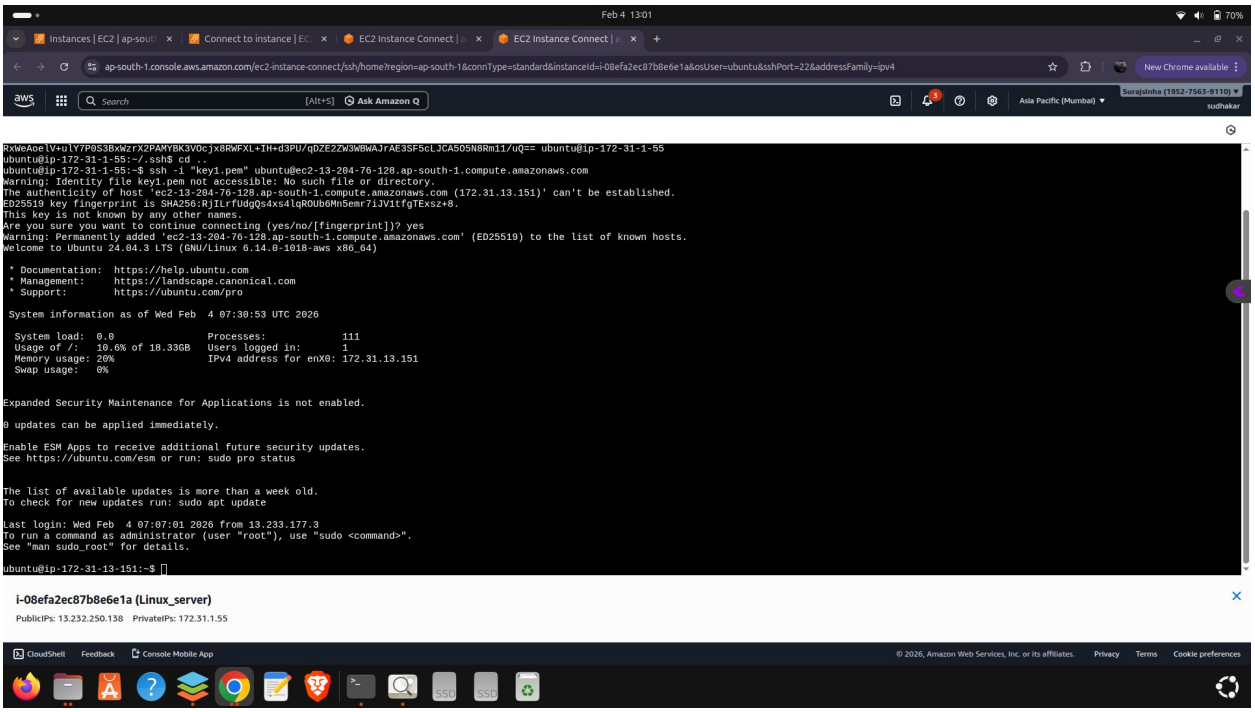


$ chmod 700 ~/.ssh

$ chmod 600 ~/.ssh/authorized_keys

$ chown -R ubuntu:ubuntu ~/.ssh


Feb 4 13:00
70%
Instances | EC2 | ap-south
Connect to instance | EC
EC2 Instance Connect |
EC2 Instance Connect |
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-011f666ae99ca7ca8&osUser=ubuntu&region=ap-south-1&sshPort=22
New Chrome available
Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
SurajsInha (1952-7563-9110)
sudhakar
ubuntu@ip-172-31-13-151:~/.ssh$ ls
authorized_keys
ubuntu@ip-172-31-13-151:~/.ssh$ vim authorized_keys
ubuntu@ip-172-31-13-151:~/.ssh$ chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
chown -R ubuntu:ubuntu ~/.ssh
i-011f666ae99ca7ca8 (Practice_server)
PublicIPs: 13.204.76.128 PrivateIPs: 172.31.13.151
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Feb 4 13:01
70%
Instances | EC2 | ap-south
Connect to instance | EC
EC2 Instance Connect |
EC2 Instance Connect |
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-08efa2ec87b8e6e1a&osUser=ubuntu&sshPort=22&addressFamily=ipv4
New Chrome available
Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
SurajsInha (1952-7563-9110)
sudhakar
RxWeAoelV+ulY7P0S3BxWzrX2PAMYBK3VOcjx8RWFXL+IH+d3PU/qDZE2ZW3WBWAJrAE3SF5cLJCA5O5N8Rm11/uQ== ubuntu@ip-172-31-1-55
ubuntu@ip-172-31-1-55:~/.ssh$ cd ..
ubuntu@ip-172-31-1-55:~$ ssh -i "key1.pem" ubuntu@ec2-13-204-76-128.ap-south-1.compute.amazonaws.com
Warning: Identity file key1.pem not accessible: No such file or directory.
The authenticity of host 'ec2-13-204-76-128.ap-south-1.compute.amazonaws.com (172.31.13.151)' can't be established.
ED25519 key fingerprint is SHA256:RjILrfUdgQs4xs4lqROUb6Mn5emr7iJV1tfgTExsz+8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-204-76-128.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Wed Feb 4 07:30:53 UTC 2026
System load: 0.0 Processes: 111
Usage of /: 10.6% of 18.33GB Users logged in: 1
Memory usage: 20% IPv4 address for enX0: 172.31.13.151
Swap usage: 0%
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Wed Feb 4 07:07:01 2026 from 13.233.177.3
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-172-31-13-151:~$
i-08efa2ec87b8e6e1a (Linux_server)
PublicIPs: 13.232.250.138 PrivateIPs: 172.31.1.55
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
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